Primary Memory, ROM Architecture & Address Mapping

Hardware Organization, Internal Array Logic, and System Address Decoding

COMPUTER ORGANIZATION & ARCHITECTURE SEMINAR

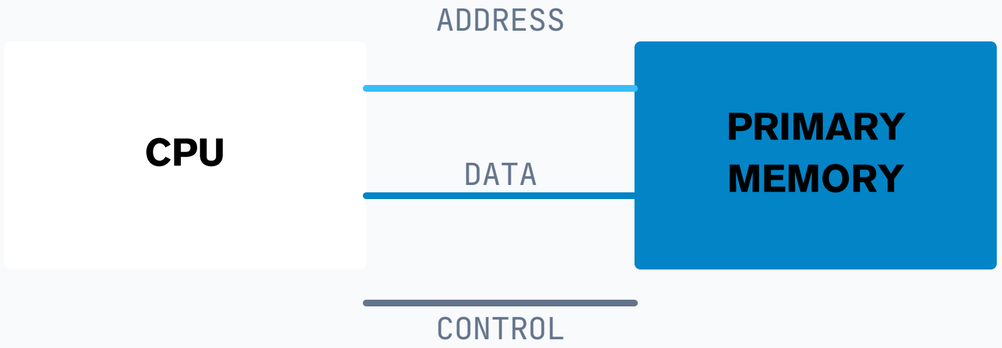
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Primary Memory Fundamentals

JOHN PAUL

Primary memory is the CPU’s immediate working store—directly coupled through address, data, and control buses. RAM prioritizes speed and volatility; ROM preserves instructions without power. Nanosecond cycle times keep access aligned with processor clocks.



VOLATILE

Fast working state
Lost without power

NON-VOLATILE

Persistent firmware
Retained without power

RAM vs ROM

JOHN PAUL

CHARACTERISTIC	RAM	ROM
Volatility	Volatile	Non-volatile
Operation	Read / write	Read-only in execution
System role	Runs OS & programs	Holds boot firmware
Control	Read + write strobes	Output enable + chip select

Classification of ROM

NISHANT

01

MASK ROM

Hardwired during fabrication
Best economics at mass scale

02

PROM

Programmed once
Fuses permanently set bits

03

EPROM

UV erasable
Floating-gate MOS cells

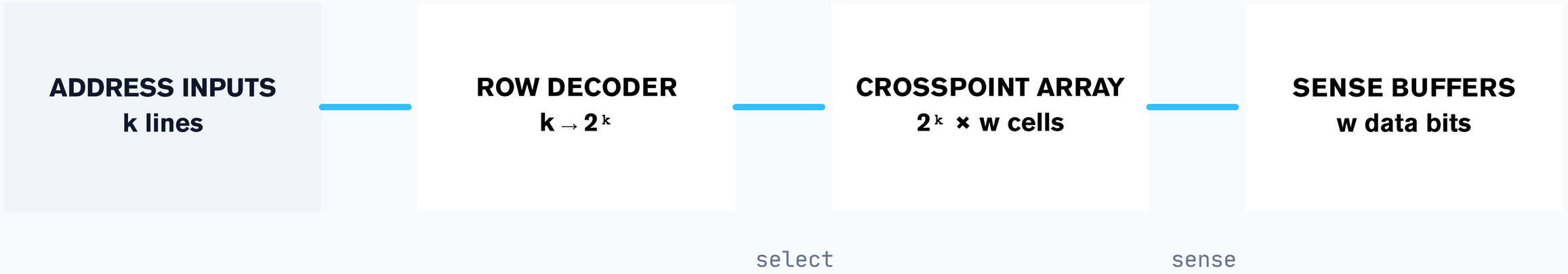
04

EEPROM

Electrically erasable
In-circuit programmable

Internal ROM Architecture

GANESH



One selected word line. w sensed output bits.

ROM Capacity Sizing

GANESH

$$\text{Capacity (bits)} = 2^k \times W$$

13 address lines

2^{13} = 8,192 words

8,192 × 8 = 65,536 bits

1

2

3

65,536 bits = 8 KB ROM

Introduction to Memory Mapping

RAM CHARAN

How the CPU selects the correct RAM or ROM location using system addresses.

DEFINITION

Memory mapping assigns specific address ranges to RAM and ROM.

CPU ACCESS

It helps the CPU locate and access the correct memory device without confusion.

UNIQUE ADDRESSING

Every individual memory location is assigned a unique system address.

ADDRESS DECODER

Decodes the incoming address bits from the CPU to identify the targeted memory chip.

CHIP SELECT ($\overline{CS}$)

Activates only the selected memory chip; all other chips remain idle.

ADDRESS MAPPING EXAMPLE

MEMORY TYPE	ADDRESS RANGE
RAM	0x0000 – 0xBFFF
ROM	0xC000 – 0xCFFF

RAM CHARAN

Address bits and control logic work together to select one memory device at a time.

HIGH-ORDER BITS

Act as chip-select control lines to choose which memory chip responds.

LOW-ORDER BITS

Act as internal address offsets to pinpoint the exact storage cell or byte within the selected chip.

ADDRESS DECODER

Dedicated combinational hardware that matches the address range to a physical chip.

CHIP SELECT ($\overline{CS}$)

An enable pin that activates data transfer for the targeted memory unit.

BUS CONTENTION

Occurs when two or more devices attempt to drive data onto the system bus at the same time. Proper address decoding and chip selection prevent this conflict.

Practical 64 KB Memory Map

SESHU

16-bit address space A_{15}–A_0 • Internal offset A_{12}–A_0

ADDRESS RANGE	BLOCK	$A_{15}A_{14}A_{13}$	SIZE
0000H–1FFFH	**BOOT ROM**	000	8 KB
2000H–3FFFH	**SYSTEM RAM 1**	001	8 KB
4000H–5FFFH	**SYSTEM RAM 2**	010	8 KB
6000H–FFFFH	**EXPANSION / RESERVED**	—	40 KB

3 high-order bits select the 8 KB block; 13 low-order bits locate the byte inside it.

Summary & Key Takeaways

SESHU

01

RETAIN

Non-volatile ROM protects boot vectors and firmware reliability.

02

DECODE

Efficient selection logic reduces pin count and hardware complexity.

03

ISOLATE

Absolute decoding prevents bus conflicts and address aliasing.

Q&A

Questions & Discussion

Thank you for your attention! The floor is now open for questions.

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